9. Introduction to Logic — P. Suppes
10. A Logical Approach to Discrete Math — D. Gries and F.B. Schneider
11. An Introduction to Discrete Mathematics, Formal System Specification, and Z: D.C. Ince
12. Programming in Prolog: W.F. Clocksin and C.S. Mellish
13. Logic, Programming and Prolog: Ulf Nilsson and Jan Maluszynski
14. Read all books by Raymond Smullyan
15. Logic: Techniques of Formal Reasoning — D. Kalish and R. Montague

1.7 Introduction to Proofs

1.7.1 Introduction

- In this section we introduce the notion of a proof and describe methods for constructing proofs. A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems. Using these ingredients and rules of inference, the final statement of the proof establishes the truth of the statement being proved.

1.7.2 Some Terminology

- Formally, a theorem is a statement that can be shown to be True.
- Less important theorems sometimes are called propositions. (Theorems can also be referred to as facts or results.)
- A proof is a valid argument that establishes the truth of a theorem.
- The statements used in a proof can include axioms (or postulates), which are statements we assume to be true (for example, the axioms for the real numbers, given in Appendix 1, and the axioms of plane geometry), the premises, if any, of the